

Green University of Bangladesh

Department of Computer Science and Engineering

CSE 403: Machine Learning

KSA 1 – Assignment

(5 Marks)

Consider the following dataset:

Age	Graduated	Income	Gender	Smokes?
18	No	12K	M	Yes
22	Yes	32K	F	Yes
27	No	59K	M	Yes
10	No	0K	F	No
29	Yes	65K	F	No
45	No	47K	F	No
59	Yes	75K	M	No
37	Yes	72K	M	No
31	No	32K	M	No
56	No	87K	M	Yes
65	Yes	54K	F	No
14	No	0K	M	Yes
39	Yes	20K	F	Yes

Task: Generate a decision tree using **Information Gain** for attribute selection.

Instructions:

- Show all calculations of gains for every attribute at every node. Just drawing the tree will not be sufficient.
- In case of a tie, prioritize the attributes in left-to-right order as shown in the table above.
- The continuous attributes **Age** and **Income** must be discretized into **3 or 4 class labels** (bins/ranges) before computing information gain. Clearly state the splitting points (ranges) you have chosen for Age and Income, along with the justification for choosing those splits.